

Topic 16: Multiple Random Variables

02-680: Essentials of Mathematics and Statistics

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Sometimes we need to deal with the interactions of more than one random variable.

Consider tossing a (fair) coin 3 times, and define two random variables:

- X — the number of heads in the first toss
- Y — the number of heads in all 3 tosses

We want to know the joint probability (that is, the probability $p(X = x, Y = y)$):

		Y			
		0	1	2	3
X	0	1/8	1/4	1/8	0
	1	0	1/8	1/4	1/8

1 Marginal Probabilities

What if we're given the joint probability of some events and want to find out the underlying probability of the events.

Example. Assume we have two unfair coins, which we will model as random variables X and Y ,

and we're given the following joint probabilities:

		Y		
		Heads	Tails	
X	Heads	1/10	2/10	3/10
	Tails	3/10	4/10	7/10
		4/10	6/10	

If we want to determine, say $P(X = \text{Tail})$, it turns out we can sum over all the possibilities of Y :

$$P(X = \text{Tail}) = \sum_{y \in \{\text{Head}, \text{Tail}\}} p(X = \text{Tail}, Y = y) = \frac{3}{10} + \frac{4}{10} = \frac{7}{10}$$

Doing all the math, both coins are biased toward **Tails**, the first coin (X) more-so.

1.1 Continuous Random Variables

For continuous random variables, this again turns from sums into integrals. Assume you're given a joint distribution function $f(x, y)$. We can, as before, find the joint probability for a range as a (double) integral:

$$p(a \leq X \leq b, c \leq Y \leq d) = \int_a^b \int_c^d f(x, y) dy dx$$

And to get a marginal distribution function

$$f(x) = \int_{-\infty}^{\infty} f(x, y) dy$$

Example. Lets say we're drawing points uniformly over the unit square, in this case

$$f(x, y) = \begin{cases} 1 & 0 \leq x, y \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

So if we want to find

$$\begin{aligned} p\left(X \leq \frac{1}{2}, Y \leq \frac{1}{2}\right) &= \int_0^{\frac{1}{2}} \int_0^{\frac{1}{2}} 1 dy dx = \left((xy) \Big|_0^{\frac{1}{2}}\right) \Big|_0^{\frac{1}{2}} = \left(x \cdot \frac{1}{2} - x \cdot 0\right) \Big|_0^{\frac{1}{2}} \\ &= \left(\frac{1}{2} \cdot \frac{1}{2} - \frac{1}{2} \cdot 0\right) - \left(0 \cdot \frac{1}{2} - 0 \cdot 0\right) = \frac{1}{4} \end{aligned}$$

2 Independence of Random Variables

Similar to independence for event probabilities, two random variables X and Y are independent iff

$$\forall x, y : p(X = x, Y = y) = p(X = x) \cdot p(Y = y)$$

Work left to the reader: one of the situations below describes independent random variables the other does not.

		Y		
		Heads	Tails	
X	Heads	1/4	1/4	1/2
	Tails	1/4	1/4	1/2
		1/2	1/2	

		Y		
		Heads	Tails	
X	Heads	1/2	0	1/2
	Tails	0	1/2	1/2
		1/2	1/2	

3 Conditional Distributions

Similar to independence for event probabilities, for two random variables X and Y :

$$p(X | Y) = \frac{p(X, Y)}{p(Y)}$$

BRCA1 mutation	P53 mutation	develops cancer	
G_1	G_2	C	$p(G_1, G_2, C)$
N	N	N	0.24
N	N	Y	0.01
N	Y	N	0.20
N	Y	Y	0.05
Y	N	N	0.15
Y	N	Y	0.10
Y	Y	N	0.05
Y	Y	Y	0.20

Compute $p(C | G_1, G_2)$.

$$p(C | G_1, G_2) = \frac{p(G_1, G_2, C)}{p(G_1, G_2)} = \frac{p(G_1, G_2, C)}{p(G_1, G_2, C = N) + p(G_1, G_2, C = Y)}$$

Note that

$$p(G_1 = a, G_2 = b) = \frac{\sum_{c \in \{Y, N\}} p(G_1 = a, G_2 = b, C = c)}{\sum_{g_1 \in \{Y, N\}} \sum_{g_2 \in \{Y, N\}} \sum_{c \in \{Y, N\}} p(G_1 = g_1, G_2 = g_2, C = c)}$$

but because the denominator is 1 it is left out of the equation above for ease exposition.

G_1	G_2	$p(C = Y G_1, G_2)$
N	N	$\frac{0.01}{0.24+0.01} = 0.04$
N	Y	$\frac{0.05}{0.20+0.05} = 0.20$
Y	N	$\frac{0.10}{0.15+0.10} = 0.40$
Y	Y	$\frac{0.20}{0.05+0.20} = 0.80$

Useful References

Wasserman. “All of Statistics: A Concise Course in Statistical Inference” §§2.5-2.7